

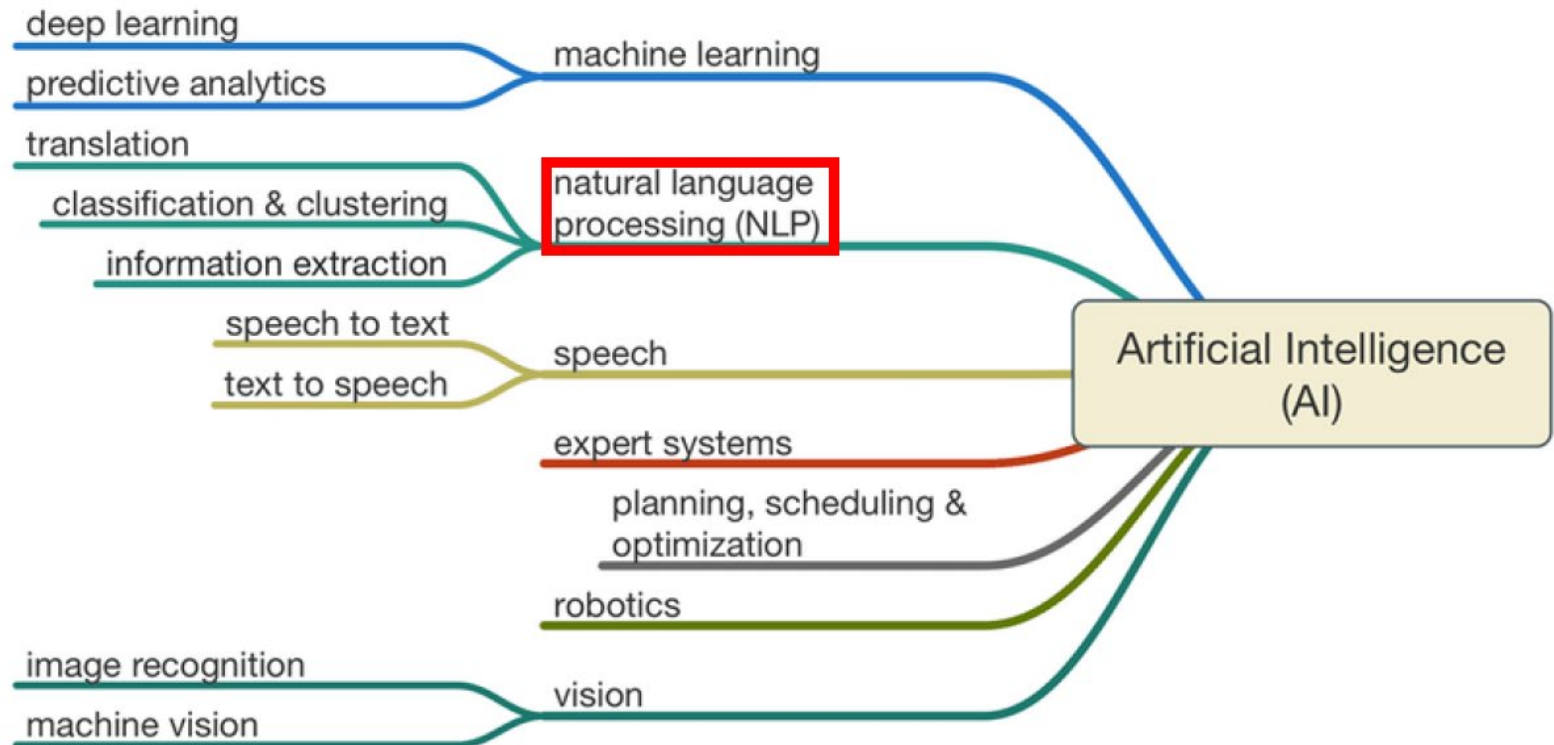
Procesamiento del lenguaje natural (NLP)

Francisco Jáñez Martino

Módulo 3

Aprendizaje Automático Aplicado

Introducción



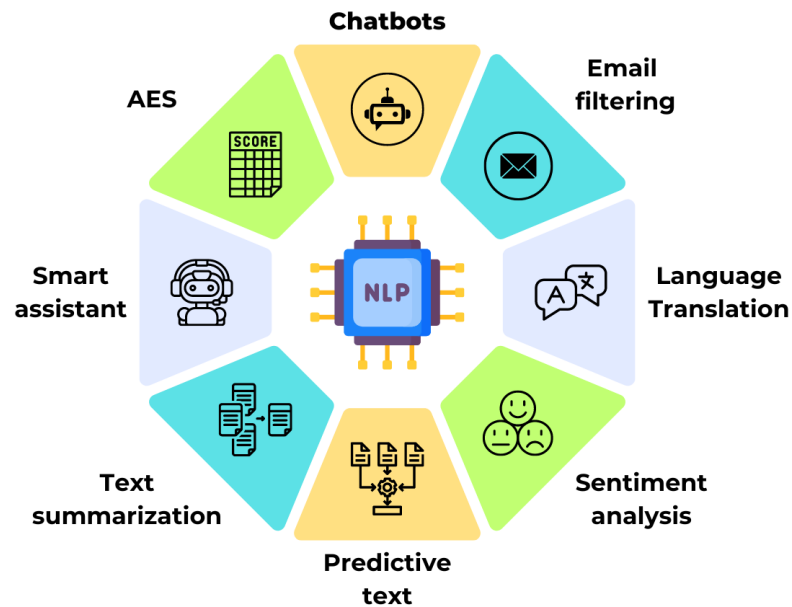
Introducción

- Tipo de dato secuencial
- Es discreto, está creado a través de una simbología y unas convenciones
- Está sujeto a cambios constantes (concept drift), nuevas palabras, formas de hablar, nuevo vocabulario
- Es composicional: varias palabras crean una oración, varias oraciones un párrafo...

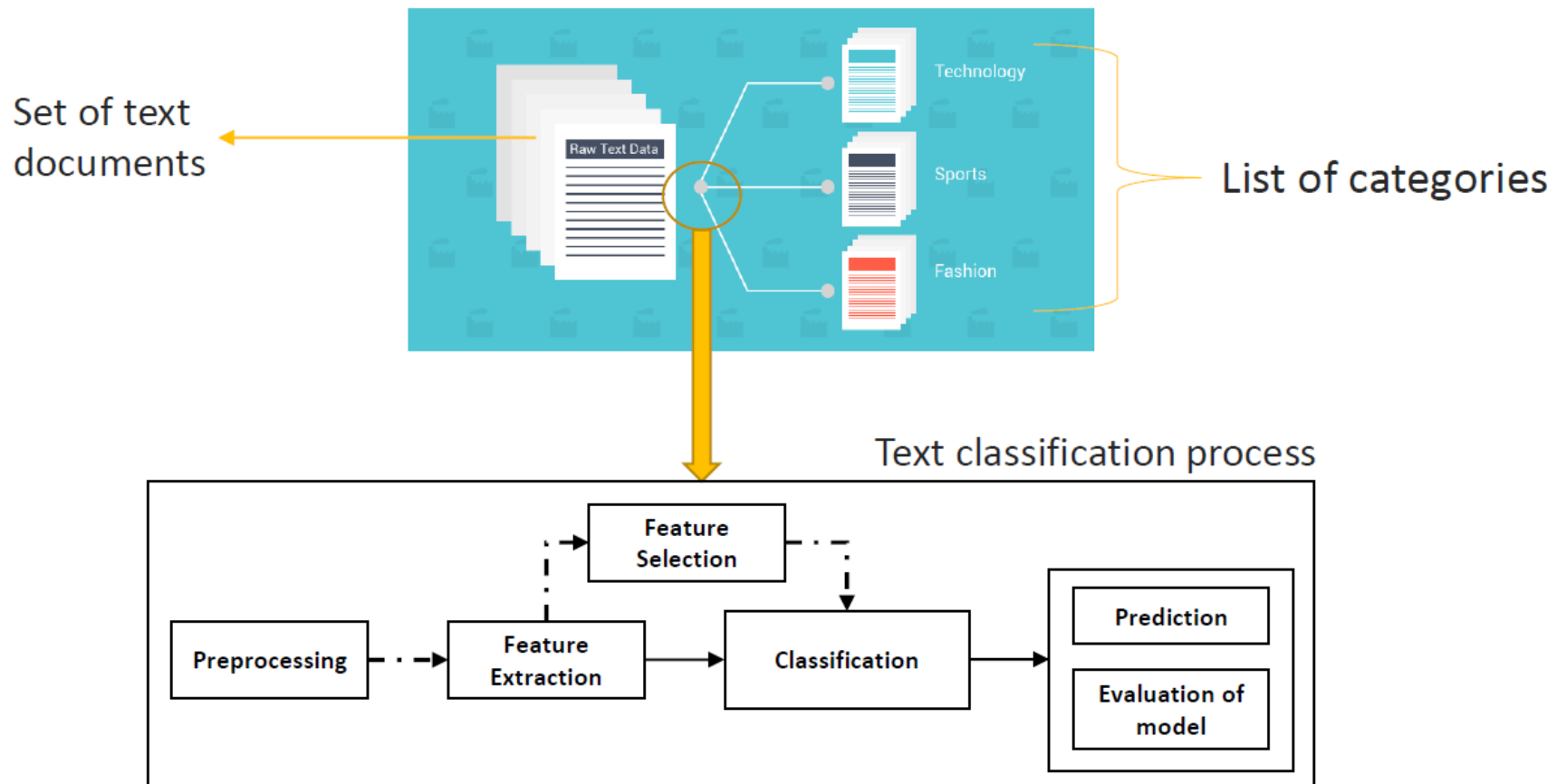
Aplicaciones



Aplicaciones



Clasificación de texto



Preprocesado

- Este proceso mejora el rendimiento de los modelos al optimizar la calidad del conjunto de datos (dataset)
- Permite limpiar el ruido, corregir errores gramaticales, reducir caracteres aislados y detectar acrónimos, entre otras tareas
- Se lleva a cabo mediante técnicas como la detección de signos de puntuación, eliminación de stopwords y reemplazos con expresiones regulares (regex)
- Su aplicación varía según el contexto y el campo específico en el que se utilice

Preprocesado



['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', 'you're', 'you've', 'you'll', 'you'd', 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', 'she's', 'her', 'hers', 'herself', 'it', 'it's', 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', 'that'll', 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'a', 'an', 'the', 'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', 'don't', 'should', 'should've', 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', 'aren't', 'couldn', 'couldn't', 'didn', 'didn't', 'doesn', 'doesn't', 'hadn', 'hadn't', 'hasn', 'hasn't', 'haven', 'haven't', 'isn', 'isn't', 'ma', 'mightn', 'mightn't', 'mustn', 'mustn't', 'needn', 'needn't', 'shan', 'shan't', 'shouldn', 'shouldn't', 'wasn', 'wasn't', 'weren', 'weren't', 'won', 'won't', 'wouldn', 'wouldn't']

Stop words list

~~We~~ sell cloned credit cards debit cards rarely



sell cloned credit cards debit cards rarely

Preprocesado

Stopwords

El análisis de datos es importante para la toma de decisiones.

Preprocesado

Stopwords

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Análisis datos importante toma decisiones

Preprocesado

Stemming

Word	Suffix	Stem
studies	-es	studi
studying	-ing	study

Lematization

Word	Morphological information	Lemma
studies	3ª person/singular verb study	study
studying	Gerund of the verb study	study

Preprocesado

Palabra	Lematización (Lema)	Stemming
Running	Run	Run
Better	Good	Better
Cats	Cat	Cat
Faster	Fast	Faster
Happily	Happy	Happi
Studies	Study	Studi
Crying	Cry	Cri
Worked	Work	Work
Relational	Relate	Relat
Mice	Mouse	Mice

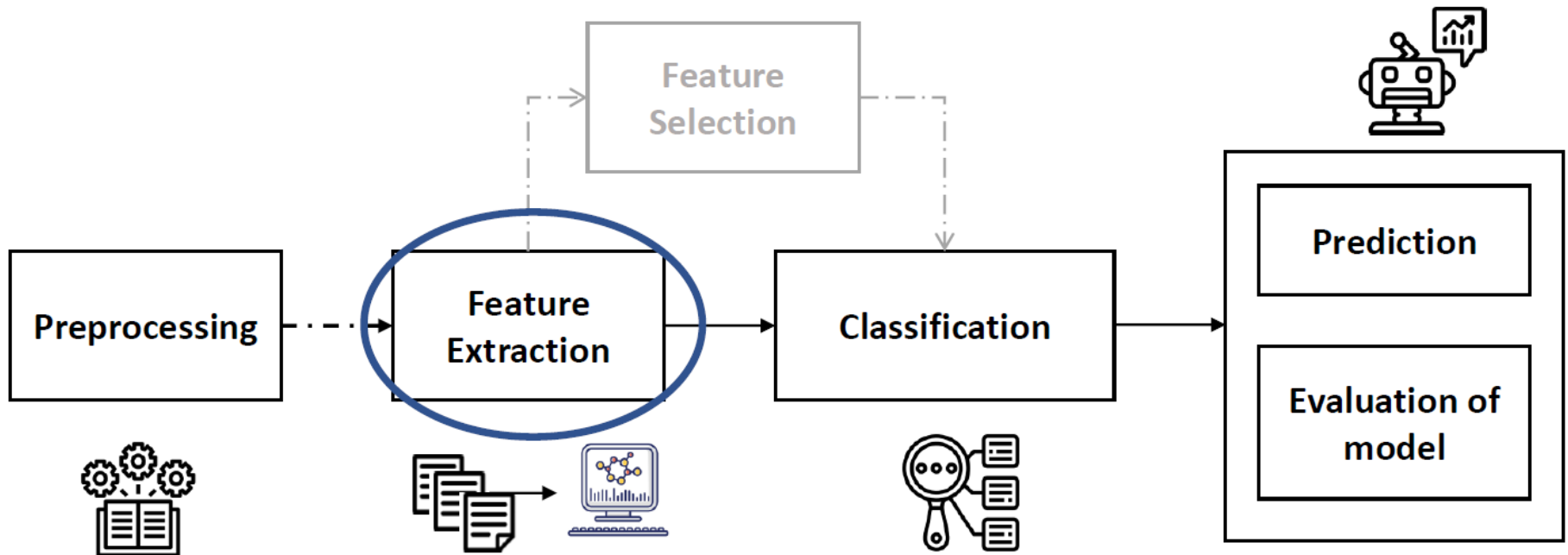


Preprocesado

- Lowercase: convertir todo a minúsculas
- Eliminar palabras duplicadas
- Reemplazo de tokens: www.unir.es [url]
- Eliminar signos de puntuación, espacios extra, caracteres individuales...

.

Extracción de características



Extracción de características BoW

	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

Extracción de características BoW

F1. Me gusta el fútbol y me gusta el baloncesto.

F2. El fútbol es un deporte emocionante.

F3. El baloncesto y el fútbol son deportes populares.

Extracción de características BoW

F1. Me gusta el fútbol y me gusta el baloncesto.

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	Me	gusta	el	fútbol	y	baloncesto	es	un	deporte	emocionante	son	deportes	populares
Frase 1	2	2	2	1	1	1	0	0	0	0	0	0	0
Frase 2	0	0	1	1	0	0	1	1	1	1	0	0	0
Frase 3	0	0	1	1	1	1	0	0	0	0	1	1	1

Extracción de características TF-IDF

- Keywords. palabras clave: aquellas que aportan información relevante para distinguir entre las distintas clases

charles bailey was indicted for feloniously stealing on the 29th of december two dressed deer skins value 20 s the property of samuel savage and richard savage richard savage i am a leather seller 63 chiswell street my partner s name is samuel savage a few days previous to the 29th of december i looked out seventy skins for an order these skins being of a bad colour i directed them to be brimstoned to make them of equal colour pale on the 29th in the afternoon i saw them all smooth on a horse a few hours afterwards they appeared very much tumbled and one was thrown into the yard and dirtied i caused them to be brought in the warehouse and counted there was two gone our foreman went to worship street and brought armstrong and vickrey they searched and found this skin in the prisoner s breeches and the other skin was found in the workshop carter i am foreman to samuel and richard savage the seventy skins i was with mr savage looking them out i took them out of the stove and counted them on the horse and on friday i counted them three times over there were no more than sixty eight instead of seventy i went to worship street brought mr armstrong and vickrey with me they waited till the men left work and when they came down they were searched and on the prisoner one skin was found john armstrong i went to this gentleman s house after the men came down vickrey and i were searching in one minute vickrey called me i received this skin from him it was taken out of the prisoner s breeches i have had it ever since john vickrey q you were with armstrong a yes while i was searching another man i saw the prisoner very uneasy and his breeches were unbuttoned i put my hand in and took that skin out he said he could not tell how it came there the property produced and identified the prisoner said nothing in his defence called four witnesses who gave him a good character guilty aged 27 confined six months in the house of correction and fined l a second middlesex jury before mr recorder

Extracción de características TF-IDF

$$TF - IDF = TF * IDF$$

The amount of times (frequency) that a term appears in a document

$$IDF_i = \log \left(1 + \frac{N_D}{f_i} \right)$$

Inverse Document Frequency for the search term i within the corpus of documents

The number of documents in the corpus of documents that contain the term D

The number of documents that contain the search term

Extracción de características

TF-IDF

Palabra	F1 (TF-IDF)	F2 (TF-IDF)	F3 (TF-IDF)
fútbol	0	0	0
baloncesto	0.0196	0	0.1761
me	0.1549	0	0
gusta	0.1549	0	0
el	0.1549	0.1549	0.1549
y	0.1549	0	0.1549
es	0	0.1549	0
un	0	0.1549	0
deporte	0	0.1761	0.1761
emocionante	0	0.6597	0
son	0	0	0.1549
deportes	0	0	0.1761
populares	0	0	0.1761



Extracción de características

TF-IDF

fútbol aparece en todos los documentos (3), entonces:

$$IDF(fútbol) = \log\left(\frac{3}{3}\right) = \log(1) = 0$$

baloncesto aparece en 2 documentos, entonces:

$$IDF(baloncesto) = \log\left(\frac{3}{2}\right) \approx 0.1761$$

deporte aparece en 2 documentos, entonces:

$$IDF(deporte) = \log\left(\frac{3}{2}\right) \approx 0.1761$$

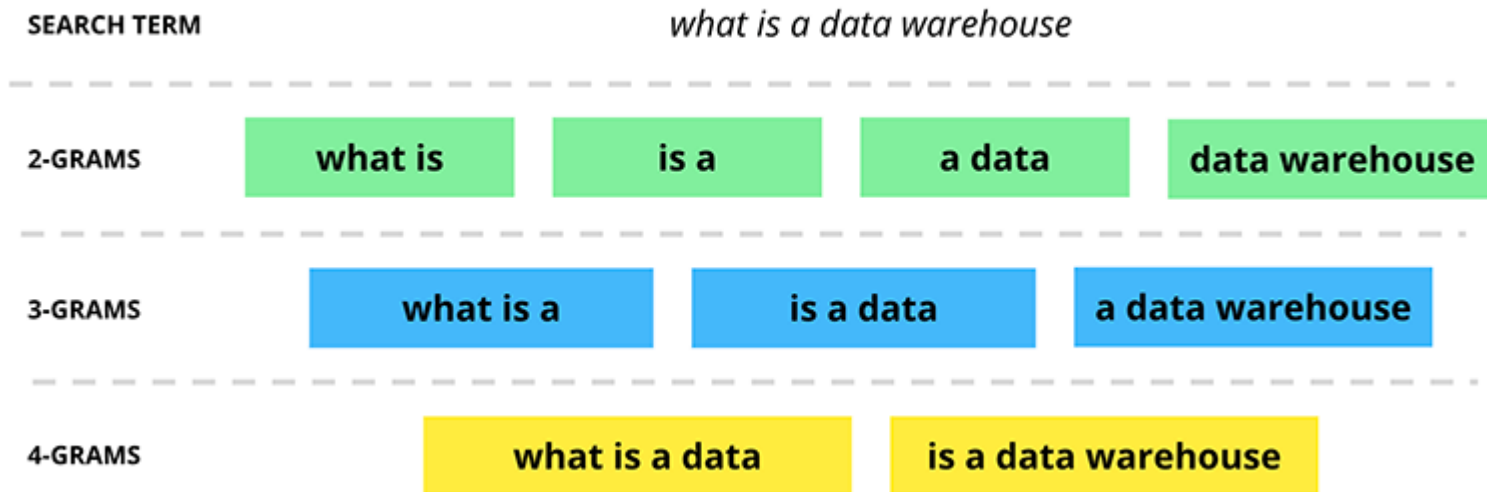
Extracción de características

Tokens y n-grams

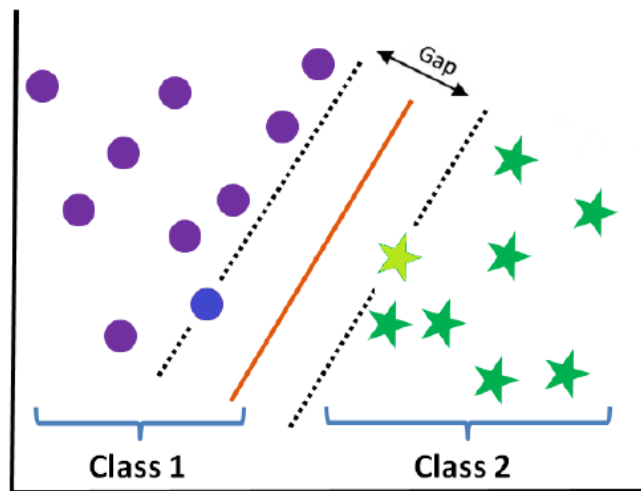
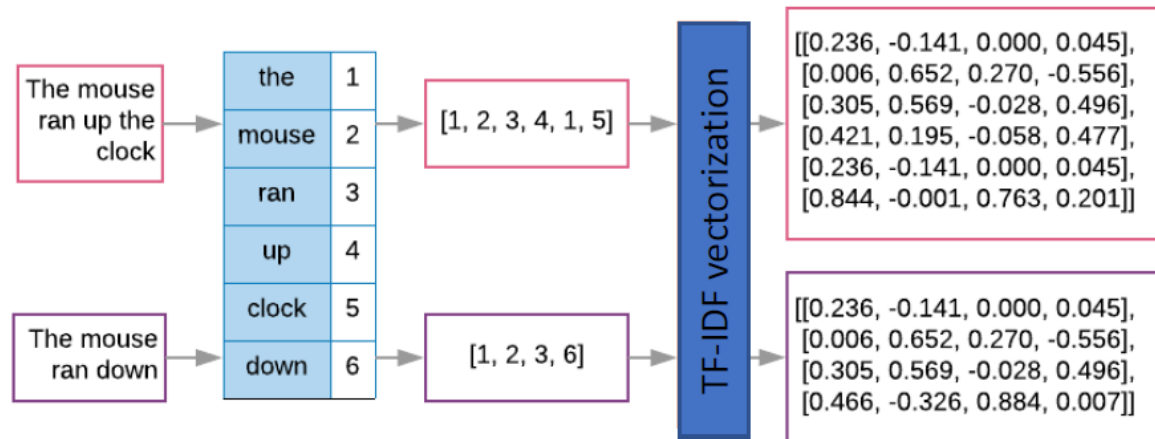


Extracción de características

Tokens y n-grams

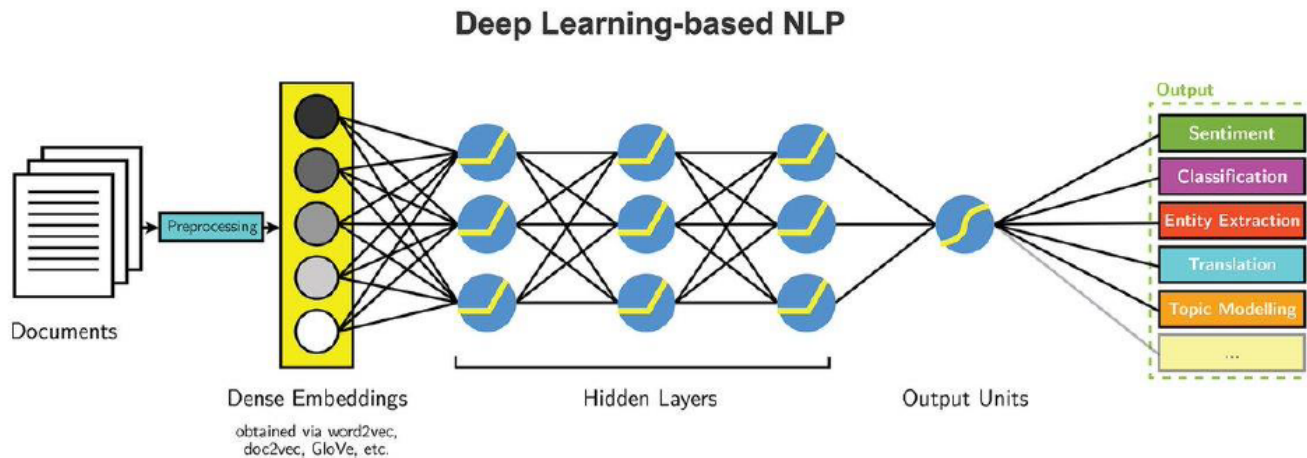
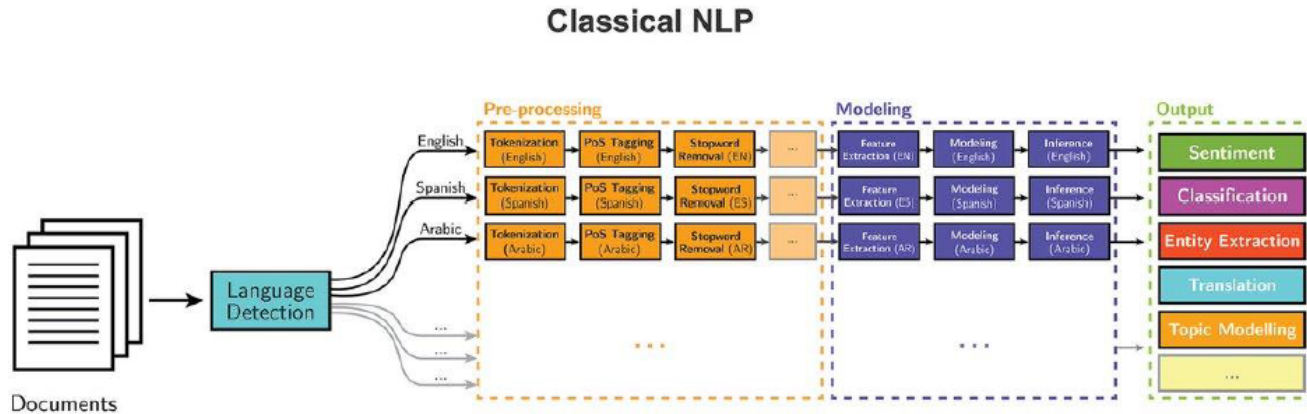


Clasificadores

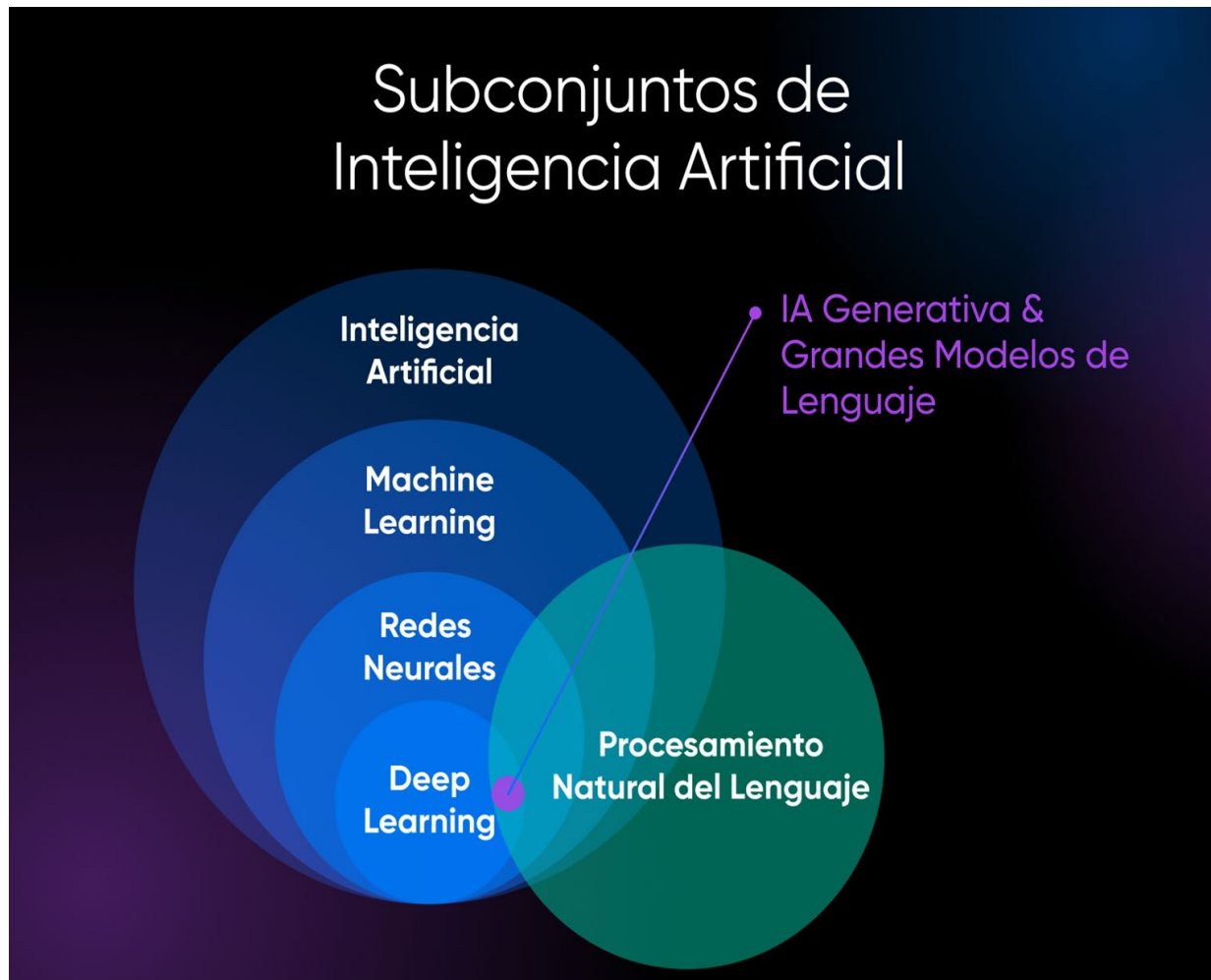


```
from sklearn.svm import LinearSVC
clf = LinearSVC(class_weight=bw, C=100, random_state=0)
```


Metodología tradicional vs aprendizaje profundo



Aplicaciones



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